

武洋

机构 新加坡科技研究局分子与细胞生物研究院 Institute of Molecular and Cell Biology (IMCB),
Agency for Science, Technology and Research (A*STAR)
地址 #05-01 Proteos Building, 61 Biopolis Drive, Singapore 138673
电话 +86-18665322610
Email wu_yang@a-star.edu.sg (机构) wuyangff95@protonmail.com (个人)
ORCID <https://orcid.org/0000-0002-1837-8330>

工作经历 / Employment

2023/02~ 至今

新加坡科技研究局 (A*STAR) 分子与细胞生物研究院 (IMCB) 博士后, 生物信息分析师

2022/08~2023/01

新加坡国立大学 (NUS) 杜克-新国大医学院 (Duke-NUS Medical School) 博士后

教育经历 / Education

2017/08~2022/07

新加坡国立大学杜克-新国大医学院计算生物学专业, 博士

2013/09~2017/06

复旦大学生命科学学院理学学士 (生物科学专业)

专业技能与项目简介 / Professional skills and brief abstract of research projects

- 生物信息学 - 基因组、转录组、健康数据的队列分析, 挖掘与疾病相关的突变位点/其他风险因素
 - 关联分析 (GWAS, TWAS)
 - 共病分析
 - 挖掘与疾病
- 生物信息学 - 肿瘤免疫微环境的挖掘
 - 单细胞转录组的挖掘
 - 空间转录组 (Stereo-seq, 10X Visium, 10X Xenium) 挖掘
 - 荧光流式细胞仪 (Cytex) 数据分析
- 肿瘤新抗原的筛选
 - PacBio Isoseq 三代测序分析结构突变、异常剪接 (Aberrant Splicing)
 - 预测新抗原的免疫原性、稳定性
 - HLA 分型 (NGS 测序、WES/WGS 数据推断: OptiType、HLA-HD 等)
 - sc-VDJ-seq 评估免疫响应、追踪 T 细胞克隆扩增、辅助 TCR-T 疗法开发
 - 免疫肽组学 - 质谱分析验证抗原是否被实际呈递
- 生物信息 workflow 构建 (Snakemake)

科研项目 / Research Projects

空间转录组学分析 / Spatial Transcriptomics

项目	核心发现	状态	相关文章
鼻咽癌免疫治疗响应 (Stere-seq)	EBV 病毒 RNA 含量可预测 Ipi+Nivo 联合治疗预后	✓ 已完成	文章 1
cDC1 抗肿瘤机制 (Stere-seq)	cDC1 耗竭导致肿瘤纤维化增加、T 细胞浸润减少	✓ 已完成	文章 4
亚洲乳腺癌 APOBEC 突变空间分布 (Stere-seq)	30kb 删减突变影响 APOBEC3A/3B 活性比例	✓ 已完成	文章 6-7
三阴乳腺癌 CD8+T 细胞抑制因子 (10X Visium)	IQGAP1、B7H3 等基因与 CD8+T 细胞丰度负相关	进行中	-

单细胞转录组学分析 / Single-cell Transcriptomics

项目	核心发现	状态	相关文章
TP53 突变与结肠癌前病变	探究 TP53 突变对结肠组织稳态及肿瘤发生的影响	进行中	-

全基因组关联研究 / GWAS

项目	核心发现	状态	相关文章
SLAS 队列老龄化研究 (NCT03405675)	克隆性造血 (如 TET2 变异) 与肝癌风险显著相关的分析	进行中	-

突变特征分析 / Mutational Signature

项目	核心贡献	状态	相关文章
COSMIC 标准突变特征库	参与发现与验证标准突变特征	✓ 已完成	文章 14
突变特征提取软件开发	开发 mSigHdp; 主导多软件性能测试	✓ 已完成	文章 10-13

发表文章 / Publications

空间转录组学与肿瘤微环境 / Spatial Transcriptomics & Tumor Microenvironment

第一作者:

1. **Wu, Y.***, Neo, Z. W., Chong, L. Y., Goh, D., Iyer, N. G., Chua, M. L. K., Lee, B., Song, X., Yeong, J. & Lim, D. W.-T. Using longitudinal spatial-omics to demonstrate in-situ Epstein-Barr virus reduction in responders to immunotherapy treated nasopharyngeal cancer. *J. Clin. Oncol.* **43**, 16_suppl (2025). https://doi.org/10.1200/JCO.2025.43.16_suppl.6025 (会议摘要 + 海报, IF: 43.4)

2. **Wu, Y.***, Wee, F., Chong, L. Y., Neo, Z. W., Lim, J., et al. Single-cell resolution spatial transcriptomics detection of pathogens followed by studying the immune milieu: using virus-associated cancers from different organs as paradigm. *J. Immunother. Cancer* **11** (2023). <https://doi.org/10.1136/jitc-2023-SITC2023.1512> (会议摘要 + 海报, IF: 10.6)

共同第一作者:

3. Wang, Y.*, **Wu, Y.***, Zhang, F.*, Periasamy, P.*, You, H., Goh, D., Fincham, R. E. A., Ning, X., Wu, D., Liu, L., Jiang, Y., Qian, Z., Yeong, J. & Zhang, Y. Integrating spatial proteogenomics in cancer research. *Adv. Sci.* (in review, 2026). (综述, IF: 17.5)

其他作者:

4. Vilbois, S., Alfei, F., Tao, H., Chuang, Y.-M., Kieffer, Y., **Wu, Y.**, Zhang, S., Chang, T.-H., Hsueh, P.-C., Wang, Y.-H., Low, J. T., Möller, S. H., Ruiz Buendía, G. A., Drobná, D., Schuster, M., Held, W., Yeong, J. P. S., Mechta-Grigoriou, F., Bénéchet, A. P., Zhou, J., & Ho, P.-C. cDC1s restrain tumor fibrosis to sustain T cell motility and T cell anti-tumor responses. *Nat. Immunol.* (审稿中, 2025). (IF: 27.6)
5. Liu, Y., Chong, C. Y. L., Wang, Y., Xu, L., Tan, Q. X., Hendrikson, J., Tan, J. W.-S., Ng, G., Guo, W., Wang, Y., Huang, L., Ge, W., Rethineswaran, V. K., Ong, J. Z. L., Quek, Y. E., Ang, A. J. Y., Teo, F., Chen, O. W., Ko, T. K., Kannan, B., Lim, B. Y., Lee, E. C. Y., Guo, Z., Lakshmanan, V., Shannon, N. B., Jegannathan, N., Ryan, C., Neo, Z. W., Chong, L. Y., Ng, A. R. H., Lee, J.-Y. J., Loo, L.-H., **Wu, Y.**, Zhang, S., Nie, L., Zhang, Q., Zeng, L., Lim, T. K. H., Ng, M. C. H., Lee, C. J. Z., Chang, P. E. J., Teh, K. K.-J., Tan, H. K., Nadarajah, R., Yong, T. T., Wang, J., Aggarwal, I. M., Fong, E. L. S., Wu, K. Z., So, J. B. Y., Yeoh, K. G., Wong, J. S. M., Seo, C. J., Cai, M., Chia, C. S., Teh, B. T., Tan, I. B., Yeong, J., Chan, J. Y., Guo, T. & Ong, C.-A. J. Identification of novel paracrine therapeutic targets in peritoneal metastases through integrative proteotranscriptomic analysis of ascites. *Cell* (审稿中, 2025).
6. Tan, Z. C., **Wu, Y.**, Neo, Z. W., Lau, M. C., Teo, S.-H., Yeong, J. P. S., Chang, S. W. & Pan, J. W. Spatial patterns of APOBEC mutagenesis in the tumour microenvironment of Asian breast cancer. *Nat. Commun.* (审稿中, 2026). bioRxiv: <https://www.biorxiv.org/content/10.64898/2026.01> (IF: 14.7)
7. Tan, Z. C., **Wu, Y.**, Neo, Z. W., Lau, M. C., Chang, S. W., Yeong, J. P. S., Teo, S.-H. & Pan, J. W. Spatial patterns of APOBEC mutagenesis in the tumor microenvironment of Asian breast cancer. *Cancer Res.* **85** (8_Suppl_1), 151 (2025). <https://doi.org/10.1158/1538-7445.AM2025-151> (会议摘要 + 海报, IF: 16.6)
8. Zhang, Y., Lee, R. Y., Tan, C. W., Guo, X., Yim, W. W., Lim, J. C., Wee, F. Y., **Wu, Y.**, Kharbanda, M., Lee, J. J., Ngo, N. T., Leow, W. Q., Loo, L. H., Lim, T. K., Sobota, R. M., Lau, M. C., Davis, M. J., & Yeong, J. Spatial omics techniques and data analysis for cancer immunotherapy applications. *Curr. Opin. Biotechnol.* **87**, 103111 (2024). <https://doi.org/10.1016/j.copbio.2024.103111> (综述, IF: 7.0)
9. Lee, R. Y., **Wu, Y.**, Goh, D., Tan, V., Ng, C. W., Lim, J. C. T., Lau, M. C., Yeong, J. P. S. Application of Artificial Intelligence to In Vitro Tumor Modeling and Characterization of the Tumor Microenvironment. *Adv. Healthc. Mater.* **12**, 2202457 (2023). <https://doi.org/10.1002/adhm.202202457> (综述, IF: 9.6)

肿瘤突变特征分析 / Mutational Signature Analysis

第一作者:

10. **Wu, Y.***, Chua, E. H. Z., Ng, A. W. T., Boot, A. & Rozen, S. G. Accuracy of mutational signature software on correlated signatures. *Sci. Rep.* **12**, 390 (2022). <https://doi.org/10.1038/s41598-021-04207-6> (IF: 4.3)

共同第一作者:

11. Liu, M.*, **Wu, Y.***, Jiang, N., Boot, A. & Rozen, S. G. mSigHdp: hierarchical Dirichlet process mixture modeling for mutational signature discovery. *NAR Genom. Bioinform.* **5**,

lqad005 (2023). <https://doi.org/10.1093/nargab/lqad005> (IF: 5.0)

其他作者:

12. Jiang, N., **Wu, Y.**, Rozen, S. G. Benchmarking 13 tools for mutational signature attribution, including a new and improved algorithm. *Brief. Bioinform.* **26**, bbaf042 (2025). <https://doi.org/10.1093/bib/bbaf042> (IF: 8.7)
13. Islam, S. M. A., Díaz-Gay, M., **Wu, Y.**, Barnes, M., Vangara, R., Bergstrom, E. N., He, Y., Vella, M., Wang, J., Teague, J. W., Clapham, P., Moody, S., Senkin, S., Li, Y. R., Riva, L., Zhang, T., Gruber, A. J., Steele, C. D., Otlu, B., Khandekar, A., ...Alexandrov, L. B. Uncovering novel mutational signatures by *de novo* extraction with SigProfilerExtractor. *Cell Genom.* **2**, 100179 (2022). <https://doi.org/10.1016/j.xgen.2022.100179> (IF: 11.1)
14. Alexandrov, L. B., Kim, J., Haradhvala, N. J., Huang, M. N., Ng, A. W. T., **Wu, Y.**, Boot, A., Covington, K. R., Gordenin, D. A., Bergstrom, E. N., Islam, S. M. A., Lopez-Bigas, N., Klimczak, L. J., McPherson, J. R., Morganella, S., Sabarinathan, R., Wheeler, D. A., Mustonen, V., Getz, G., Rozen, S. G., Stratton, M. R. & P. Consortium. The repertoire of mutational signatures in human cancer. *Nature* **578**, 94–101 (2020). <https://doi.org/10.1038/s41586-020-1943-3> (IF: 55.0)

蛋白质组学 / Proteomics

15. Periasamy, P., Soon, K., Lim, X., **Wu, Y.**, Meng, J., Lim, F., Yeong, J. DETECT – Cost-effective, high-throughput enrichment of low molecular weight proteins in serum proteomics – insights from a CAR-T therapy trial. *J. Immunother. Cancer* **11** (2023). <https://doi.org/10.1136/jitc-2023-SITC2023.0200> (会议摘要 + 海报, IF: 10.6)

生物信息学工具开发 / Bioinformatics Tool Development

16. Li, Z., Zhang, F., Wang, Y., Qiu, Y., **Wu, Y.**, Lu, Y., Yang, L., Qu, W. J., Wang, H., Zhou, W. & Tian, W. PhenoPro: a novel toolkit for assisting in the diagnosis of Mendelian disease. *Bioinformatics* **35**, 3559–3566 (2019). <https://doi.org/10.1093/bioinformatics/btz100> (IF: 7.1)